

70 Billion Parameters vs 8 Billion Parameters

70 billion parameters vs 8 billion parameters means the number of learnable values (parameters) inside an AI model. These parameters help the model learn patterns, relationships, and information during training.

Simple Example

Think of an AI model like a brain:

- **70B model** → Bigger brain → Can learn and handle more complex patterns
- **8B model** → Smaller brain → Fewer parameters, but generally works faster

Feature	70 Billion	8 Billion
Parameters	70,000,000,000	8,000,000,000
Model size	Larger	Smaller
Understanding	Generally higher	Good
Complex tasks	★★★★★	★★★★
Speed	Generally slower	Very fast
Memory/compute	Requires more	Requires less
Cost	Generally higher	Generally lower

What Are Parameters?

A **parameter** is a numerical value that the model learns during training.

For example, when a model reads many news articles, it learns:

Words → Patterns → Relationships → Knowledge Representation

Parameters are used to store what the model learns from these patterns.

In Your Project

Your project workflow is:

News Article → AI Summary → Tamil Translation

A **70B model** generally has more capacity to:

- Understand long news articles
- Identify important information
- Generate better summaries
- Maintain context during Tamil translation

An 8B model

- Can perform the same tasks

- May have lower quality on very complex articles or contexts compared with a larger model
- Can operate **faster and with fewer computing resources**

Very Simple Explanation

70B = Larger + More capability + More resources

8B = Smaller + Faster + Fewer resources

■■ However, **70B does NOT mean it is always 8.75 times “smarter” than 8B**. The number of parameters is only one factor used to evaluate an AI model's capability.